Blackmon Lab

Evolutionary Biology

Genome Size as a Driver of Chromosomal Volatility in Chrysomelidae

Ava Tingue & Heath Blackmon
Department of Biology

TEXAS A&M UNIVERSITY Biology

Background

The diversity of chromosome numbers (karyotypes) varies wildly across the tree of life, yet the forces driving this variation remain poorly understood. The Drift Barrier Hypothesis suggests that as genome size increases, the efficiency of natural selection decreases. In these "large" genomes, chromosomal rearrangements like fissions and fusions may not be as effectively purged, leading to an increased rate of change. We tested this by examining the relationship between C-value (genome size) and haploid chromosome number in the beetle family Chrysomelidae.

Study Design & AI

This study utilized a comparative phylogenetic framework to analyze 24 species within the family Chrysomelidae for which both time-calibrated genomic data and karyological records were available. We integrated data from the Animal Genome Size database and published karyotype records, ensuring taxonomic consistency across all datasets. To account for the non-independence of species data, we utilized a high-resolution chronogram as the backbone for all evolutionary modeling. A novel component of our workflow involved the integration of Large Language Models (LLMs) to optimize the bioinformatics pipeline. AI was employed as a collaborative tool for R script troubleshooting, data structural alignment, and the development of custom R-functions to visualize complex stochastic processes. This integration allowed for a more agile and error-resistant transition from raw data to sophisticated phylogenetic modeling.

Model Fitting

I employed Phylogenetic Generalized Least Squares (PGLS) to test for a relationship between genome size and chromosome number. Under a Brownian Motion correlation structure, the model revealed no linear correlation ($p = 0.87$), confirming that genome expansion does not drive directional change. Instead, this lack of linearity, paired with the observed heteroscedasticity observed in my boxplot analysis, suggests a "relaxation" of chromosomal constraints. To map this trajectory, we performed Ancestral State Reconstruction (ASR) via maximum likelihood estimation (Figure 5). Finally, we quantified karyotypic velocity through 100 iterations of Stochastic Character Mapping, estimating the total fissions and fusions throughout the clade's history. This multi-model approach provides a robust statistical foundation for the drift-barrier hypothesis in Chrysomelidae.

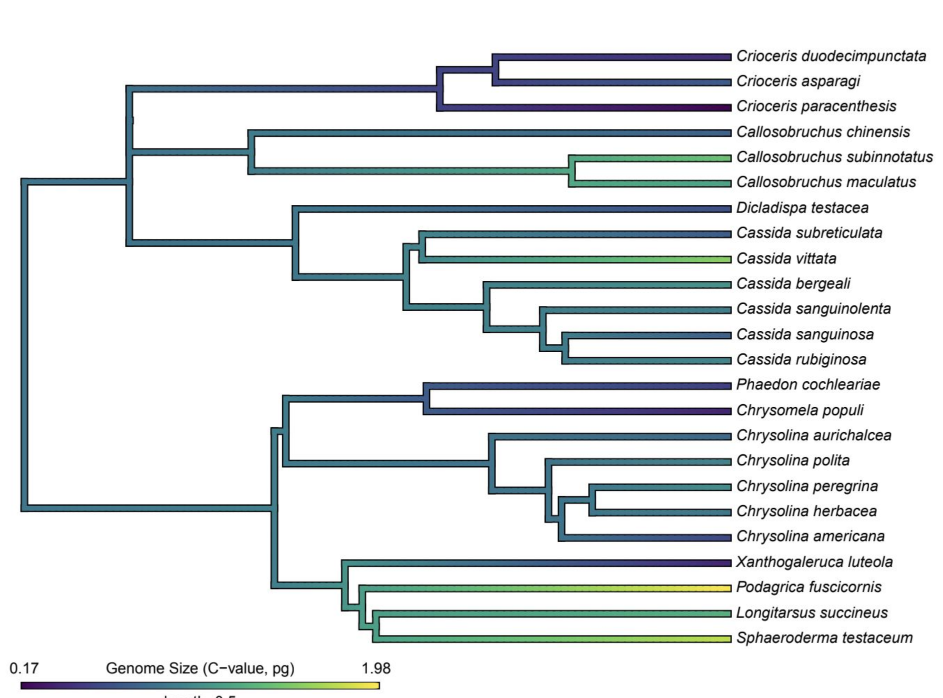


Figure 5. Ancestral State Reconstruction of Genome Size. Visualization of the evolution of C-values across the Chrysomelidae phylogeny. The color gradient on branches represents the estimated ancestral genome size (pg), illustrating shifts in DNA content across different lineages.

Results

Figure 1. Chronogram and Karyotype Evolution of Chrysomelidae. A time-calibrated phylogeny of 24 leaf beetle species showing the relationship between genome size (indicated by colored tip nodes) and haploid chromosome numbers (n), represented by the blue barplot.

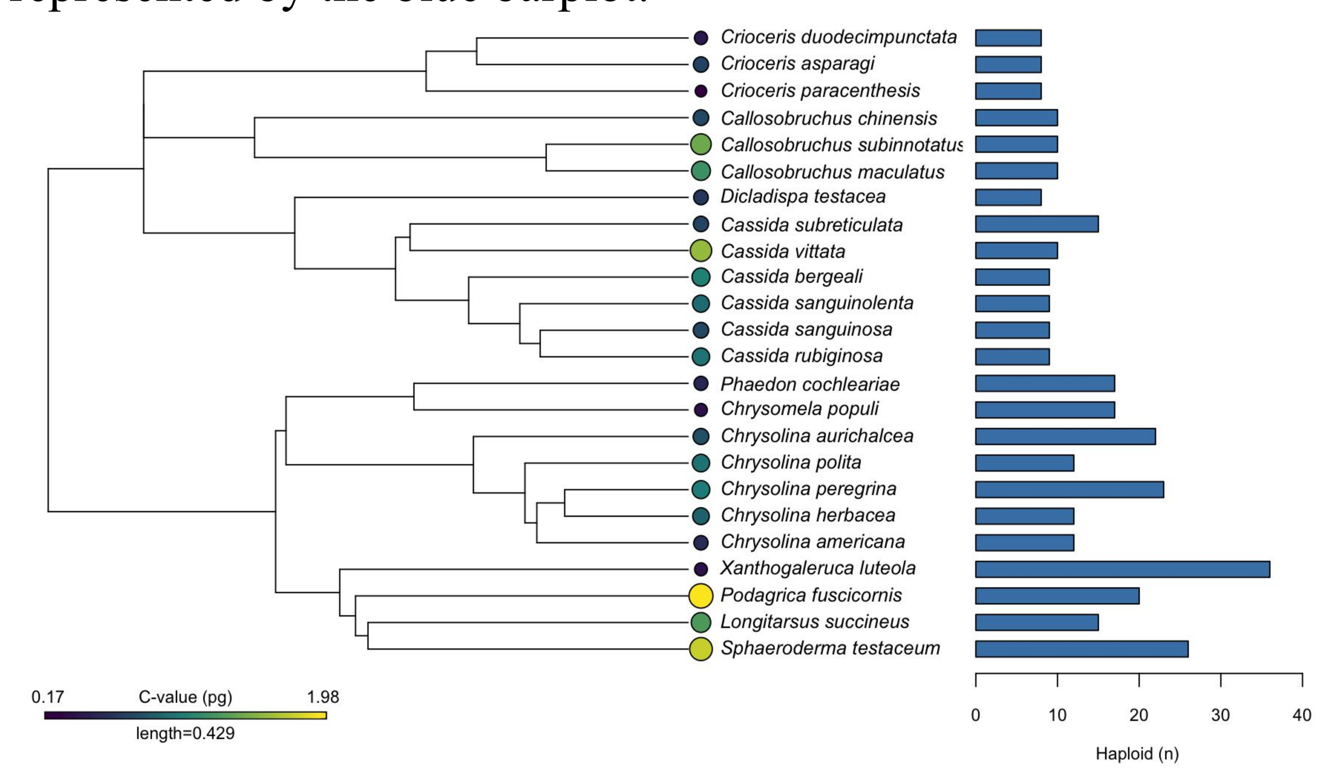


Figure 2. ChromePlus model for the evolution of chromosome number. At an instance in time a lineage will have i chromosome and either binary state 1 or binary state 2. A lineage can make four possible transitions: δ the fusion of two chromosomes, γ the fission of a chromosome, ρ a whole genome duplication, and a transition in centromere type (i.e. transition from state 1 to state 2 q_{12}, or transition from state 2 to state 1 q_{21}.

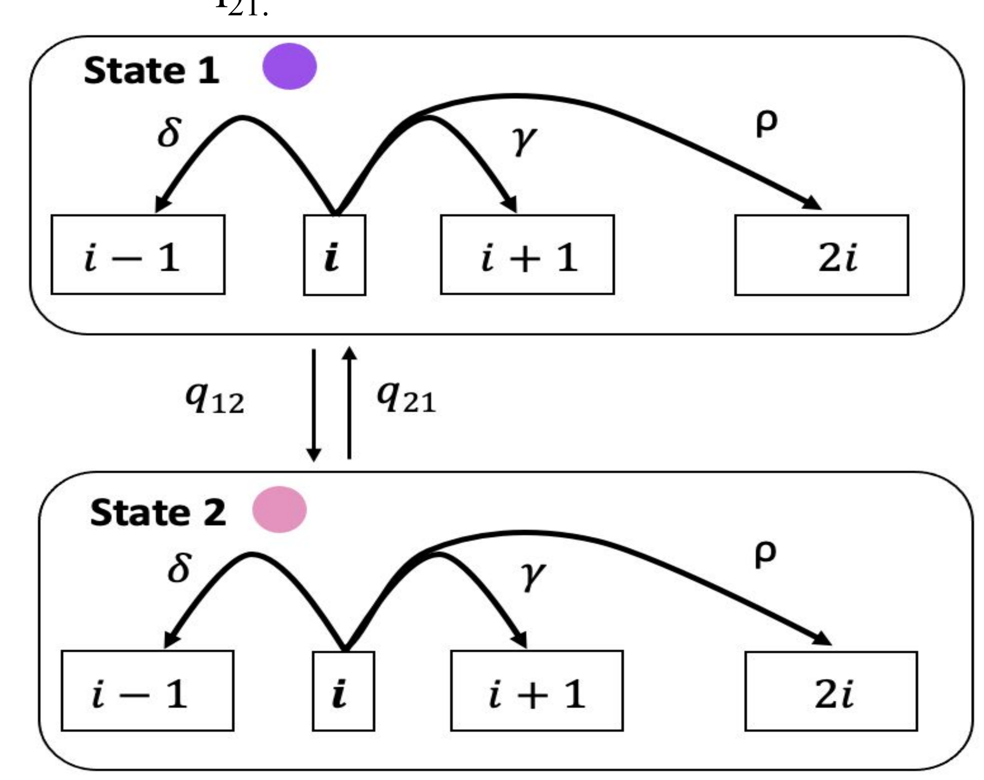


Figure 3. Relationship Between Genome Size and Chromosomal Variance. The boxplot compares chromosome number distributions across genome size categories. Species with "Large" genomes (>0.5 pg) exhibit significantly higher variance in haploid numbers compared to those with "Small" genomes, supporting the hypothesis that increased DNA content reduces selective pressure against chromosomal rearrangements.

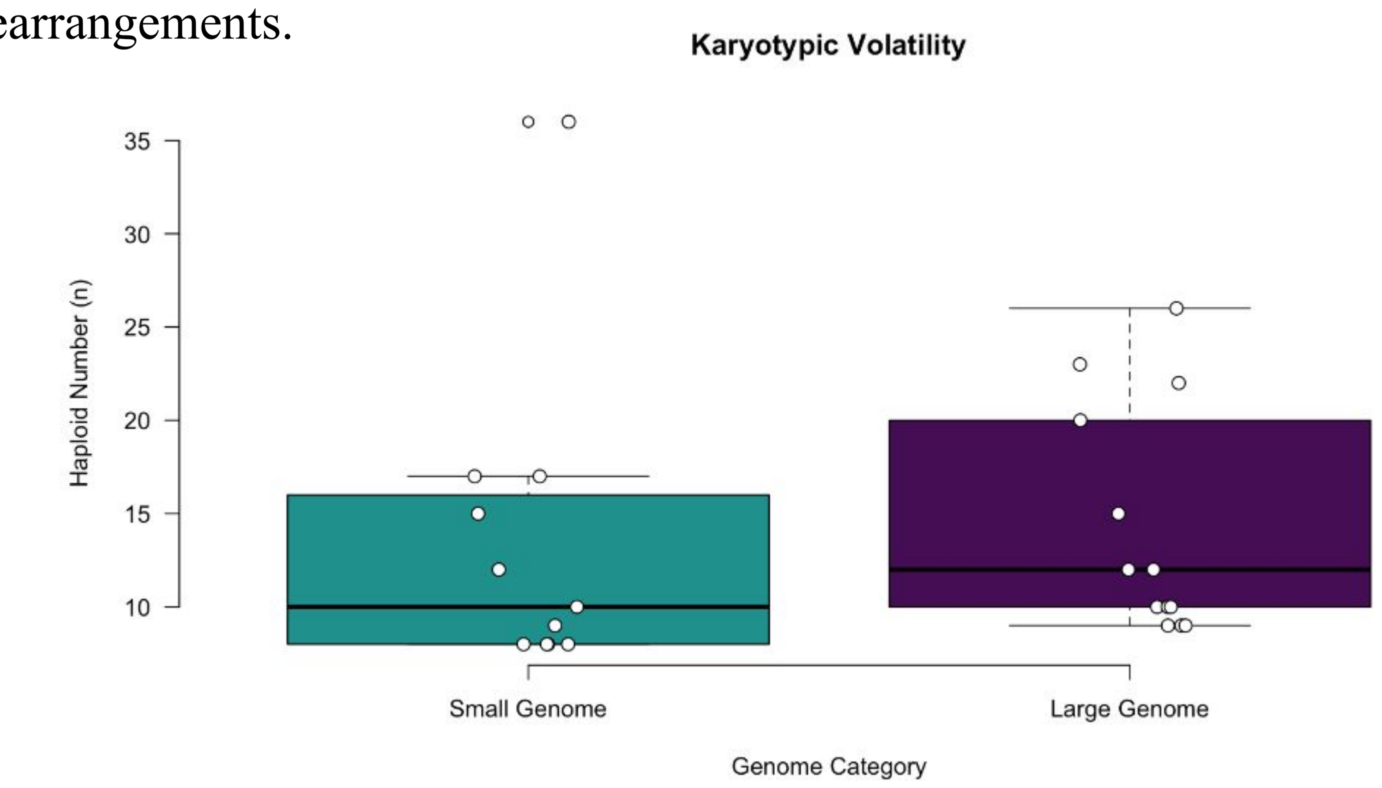
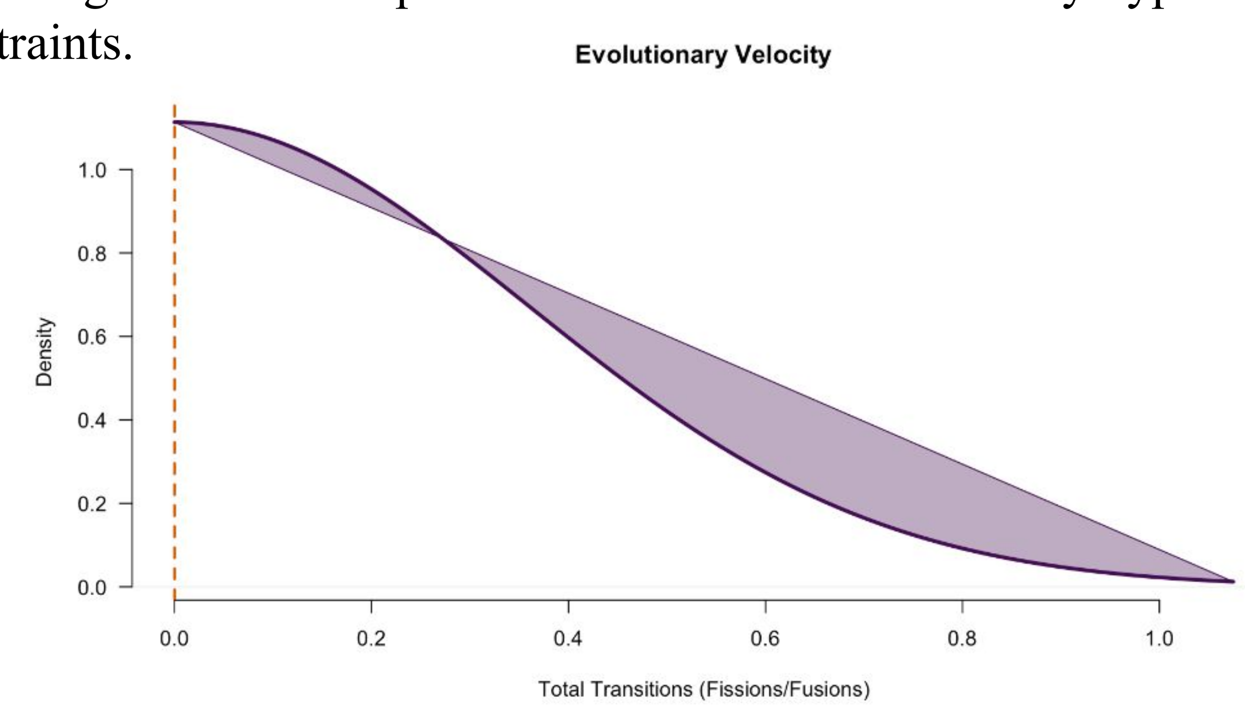


Figure 4. Evolutionary Velocity of Chromosomal Traits. The right-skewed distribution in the evolutionary velocity analysis suggests that while karyotypic stasis is common across the clade, a subset of lineages exhibits high rates of chromosomal transition. This increased velocity aligns with our finding that larger genomes are associated with significantly higher chromosomal variance (Figure 3), suggesting a link between genome size expansion and the relaxation of karyotypic constraints.



Discussion

Our findings support the idea that genome size acts as a mediator of chromosomal stability. In lineages with small genomes, selection appears to maintain a 'tight' karyotype around n=10-12, likely due to the high fitness costs of rearrangements in streamlined genomes. However, in large-genome lineages, the increased variance suggests that the 'brakes' of natural selection have been released. This relaxation allows both fissions and fusions to occur more frequently, driven by an accumulation of repetitive elements that facilitate structural errors. Our simulation of evolutionary velocity confirms this instability, showing a rapid rate of transition that characterizes the family. Ultimately, genome expansion creates a permissive environment for karyotypic flux, shifting the evolutionary trajectory from stability to volatility.

Conclusions & Acknowledgements

Increased genome size correlates with high chromosomal volatility, supporting the Drift Barrier Hypothesis in leaf beetles.

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